

AI Research Agent

Functional portfolio demonstration • Make.com + Supabase + HTTP + Groq

Project status

This is a functional portfolio demonstration built and tested with demonstration data. It is not a live client deployment and does not claim measured client results.

1. Project overview

The project automates repeated company research through three connected Make.com scenarios. Website content is retrieved, analysed with Groq, checked by a second AI call for specificity and usefulness, and only approved analyses continue to a final structured report.

Business problem	Repeated company research is slow, inconsistent and can produce generic AI output that sounds useful but is not specific enough to act on.
Solution	A three-scenario Make.com pipeline with Supabase state management, HTTP website retrieval, Groq analysis, a separate quality-control step and final report generation.
Tools	Make.com, Supabase, HTTP module and Groq.
My role	Planning, workflow design, build, testing and documentation for the portfolio demonstration.

Scope

- In scope: company queue, website retrieval, AI company analysis, status locking, separate quality evaluation, approved/needs-improvement routing, final master report and Supabase status management.
- Out of scope: multi-source search, source citations, automatic retry/error queue, mandatory human approval and live client deployment.

2. Implemented workflow

The implemented solution is split into three scenarios. Supabase statuses connect the scenarios and also act as simple workflow locks so the same company is less likely to be processed twice.

Scenario 1 - Company Analysis

1	Searches Supabase for a company with status queued.
2	Immediately changes the status to processing before other work starts.
3	The HTTP module retrieves content from the company website.
4	Groq receives the website content and a structured prompt, then returns a company analysis as JSON.
5	The analysis is saved to Supabase and status changes to analyzed.

Scenario 2 - Quality Check

1	Finds a record with status analyzed and changes it to quality_checking.
2	A second Groq call evaluates whether the analysis is specific and useful or too generic.
3	Groq returns structured JSON. The Make.com router reads that value and chooses the route.
4	Approved -> status becomes approved and Scenario 3 can continue.
5	Needs improvement -> feedback is saved and status becomes needs_improvement; the record stops before the final report.

Scenario 3 - Final Master Report

1	Finds a record with status approved.
2	Changes the status to report_generating.
3	Groq creates the final structured master report from the approved analysis.
4	The report is saved to Supabase and status changes to final.

Important architecture note

The AI does not decide its own next step. Groq returns structured output, while Make.com uses deterministic routing rules to decide what happens next. This is AI automation, not an autonomous agent.

3. Quality-control logic

- Too generic: repeats common website language, lacks company-specific observations, gives nothing concrete to act on, or does not follow the expected structure.
- Ready to approve: uses specific website information, includes useful company-specific observations, follows the structure and is clear enough for the final report.
- The quality check reduces weak output, but it does not guarantee factual accuracy. Human review is still recommended before important business use.

4. Workflow states

queued	Waiting for Scenario 1.
processing	Scenario 1 has locked the record and is retrieving/analyzing content.
analyzed	Company analysis saved; ready for Scenario 2.
quality_checking	Scenario 2 is evaluating the analysis.
approved	Quality check passed; ready for Scenario 3.
needs_improvement	Feedback saved; does not continue to final report.
report_generating	Scenario 3 is creating the final report.
final	Final master report saved; pipeline complete.

5. Testing and acceptance criteria

The scenarios were tested with demonstration data. These criteria describe what the portfolio build is intended to prove; they are not measured client results.

- A queued company is picked up only by Scenario 1.
- Status is changed before the longer processing steps begin.
- Website content is passed to Groq through the HTTP module.
- An analysis marked needs_improvement does not continue to the final report.
- An approved analysis is picked up by Scenario 3 and saved as a final report.
- The main status transitions and routing branches are visible in Supabase/Make.com.

6. Risks and current limitations

Risk	What could happen	Priority	Mitigation / next step
Weak or incorrect AI conclusion	The report can sound confident while being wrong.	High	Human review and source references before important use.
Website fetch failure	The HTTP module may receive incomplete or blocked content.	High	Add retry logic, fallback handling and clearer error states.
Generic analysis	The result is too broad to be useful.	High	Separate quality-check scenario and needs_improvement route.
Duplicate processing	Two runs may pick up the same company in edge cases.	Medium	Current status lock reduces the risk; stronger transactional locking would be better.
Malformed AI JSON	Router logic can fail if the output format is invalid.	Medium	Validate structured output and add error handling.
No source citations	Claims are harder to verify.	Medium	Planned V2: save source URLs/titles and link claims to sources.

Additional limitations

- The current version mainly uses the company website rather than multiple independent sources.
- The quality check evaluates specificity and usefulness, not factual truth.
- There is no automatic retry/error queue for failed HTTP or Groq calls.
- There is no mandatory human approval step before status final.
- The workflow is a portfolio demonstration, not a production deployment.

7. Planned improvements

- Add multi-source research with a search API alongside the official website.
- Save source title, URL, date and source type for traceability.
- Connect important report claims to supporting sources.
- Flag unsupported AI conclusions and conflicting sources.
- Add retry and error handling for HTTP and Groq failures.

- Add a human review step before the final report is marked complete.
- Add monitoring for stuck or delayed records.

8. Interview quick recap

What did you build?	A three-scenario AI research workflow in Make.com for repeated company research.
Why three scenarios?	To separate analysis, quality control and final reporting, with Supabase status fields handing records from one scenario to the next.
What is the key differentiator?	A separate quality-check layer that stops generic output before the final report.
Who makes the routing decision?	Make.com. Groq returns structured JSON; the router reads the value and follows a fixed branch.
Why Supabase?	It stores records, analyses, feedback, reports and the workflow status between scenarios.
Biggest limitation?	No source citations or multi-source verification yet, and AI output still requires human review.